LESSON 129 (0.5 GROUND and 2.0 DUAL CROSS COUNTRY)

Lesson Objective

This lesson is used to improve the student's ability to plan and fly a cross-country flight to a non-towered airport more than 50 NM from the original point of departure. This lesson will also increase the student's ability to navigate using pilotage, dead reckoning and radio navigation.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
    - Review any deficiencies from lesson 128.
    - Plan a cross country for lesson 129: KMLB-KSUA-KMLB, KMLB-KOBE-KMLB, or another suitable non-towered airport.
- Student questions.

References

- Advisory Circulars. (Search Tool)
    - Aviation Weather. (AC 00-6B)
    - Aviation Weather Services. (AC 00-45H)
    - Pilot's Guide to a Preflight Briefing. (AC 91-92)
- Aeronautical Information Manual.
    - Chapter 01 - Air Navigation.
    - Chapter 03 - Airspace.
    - Chapter 06 - Emergency Procedures.
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
    - Chapter 16 - Navigation.

Lesson Content (Ground)

Review Cross-Country Flight Planning

- Airspace
    - Review the student's chosen route and verify the student's consideration regarding airspace.
    - Cruising altitudes should be appropriate for the airspace.
- Weight and Balance
    - Review the student's weight and balance planning on the TOLD card.
    - Verify that the fuel information on the TOLD card matches the fuel information on the navlog.
- Performance and Limitations
    - Review the student's calculations for pressure altitude and density altitude.
    - Verify that the V speeds were filled out correctly on the TOLD card.
    - Inspect the totals (times, fuel, distances).

- Navigation Log
  - Verify the entirety of the nav log.
  - Cruising altitudes should be appropriate for the magnetic course.
  - Ensure that the airport diagrams are drawn, and the frequencies are filled in.
  - Verify that the student has the line drawn and checkpoints identified in the sectional chart.
  - Ensure that the student has TOC and TOD calculations complete.
  - The information on the fuel management table must match the information on the front of the nav log.
  - Verify that the flight plan portion has the required and correct entries.
  - VOR information should be written.
- Weather
  - Review the student's weather planning.
  - Ensure that the student obtained a standard weather briefing.
  - Ask the student questions about any potential weather hazards for the route of flight.
  - Ensure that the student has filled out the weather section of the navlog.
  - Cruising altitudes should be appropriate for the weather.
- Flight Plan Filed with FSS
  - Ensure that the student has filed the flight plan with FSS.
  - Verify departure, enroute, and arrival times are correct.
- Brief the student:
  - Opening the flight plan.
  - Getting flight following.
  - Diversion procedures with the student as this will be performed during this cross-country.

Cockpit Management

- Brief management of radio frequencies.
- Advise how to keep resources orderly.
- Emphasize the need to think ahead of the airplane.

Go/no-go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)

    - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

<u>**Lesson Content (Dual Cross Country)**</u>

Pilotage and Dead Reckoning

- Departure:
    - Set up radios, GPS, Navaids, etc...
    - TIME OFF must be recorded.
    - Radio Communications:
        - Open flight plan
        - Receive flight following.
        - Coach student if needed.
- Cruise:
    - Make sure the student is switching tanks and leaning.
    - Evaluate CRM and division of attention.
    - Pilotage & Dead Reckoning
        - The student looks for checkpoints.
        - Course must be intercepted.
        - ATAs recorded.
        - Record actual Ground Speed.
        - Calculate revised ETA.
        - Verify location.
        - Complete the nav log.
    - Radio Communications
        - Coach student if needed.
- Arrival:
    - Use proper TOD.
    - Complete checklists.
    - Radio Communications:
        - Advise when to check the weather.
        - Cancel flight following.
        - Ask the student to close the flight plan.
    - Completing Navigation Log
        - Complete last time box when safe.

Completing Navigation Log

- The nav log must be completed upon reaching the checkpoints.
- This includes:
    - Groundspeed.
    - Time/ATA.

    - Fuel burn.
- If the student is task saturated while crossing a checkpoint, remember:
    - Aviate, navigate, communicate!
    - Complete the nav log once time permits.

Flight Plan Usage (opening/closing)

- This was demonstrated to the student on the first cross-country.
- Have the student speak to FSS to open and close the flight plan.
- If necessary, coach the student.

Emergency Procedure (Oil pressure)

- On the return leg, simulate a low oil pressure scenario.
- Ensure that the student can run through the checklist from memory.
- Verify with the checklist if time permits.
- Question the student in their decision making.
- What can low oil pressure lead to?
- Can it lead to a power-off landing?

Diversion Procedure

- Simulate a scenario where a diversion is necessary (unforecasted weather, system malfunction, instrument malfunction, engine complications, lost comms, etc.).
- The student must confirm the present position on the sectional chart.
- Choose an alternate shown on the sectional or use the 'Nearest' page on the GPS.
- The student must divert immediately toward the alternate using shortcuts/rule of thumb calculations (above) if it is an emergency. If time permits, the student may circle until all of the information is obtained (heading, time, frequencies, etc.)
- Diversion log must be filled out.
- The chart supplement must be used to obtain the required information for the alternate airport.
- Stress aircraft control.
- The student must show situational awareness.
- Flight plan should be closed.

Lost Procedures

- While departing, on the second leg, direct the student to an unfamiliar area.
- Take away the student's sectional charts, GPS course, and radio navigation.
- Present the lost scenario.
- Give the student the charts back, access to the GPS, and radio navigation.
- Verify that the student says and does the correct lost procedure (7C's).

Flight Following

- Have the student request flight following on both legs.
- If necessary, help the student.

Short-Field takeoff and Maximum Performance Climb

- <u>Student does, student says:</u>

- Verify correct procedure.
- Ensure that the student holds the brakes until full static RPM.
- Verify that the climb out is conducted at 63 knots.
- Verify that the student says "obstacle cleared" if there is an obstacle or simulated obstacle.
- Verify that the student says positive rate of climb before retracting the flaps.

Short-Field Approach and Landing

- Student does, student says:
  - Verify correct procedure.
  - Emphasize the importance of not forcing the airplane on the ground. If the point cannot be made, perform a go-around.
  - Verify simulated maximum breaking.

Soft-Field Takeoff and Climb

- Student does, student says:
  - Verify correct procedure.
  - Verify that the climb out is conducted at 63 or 79 knots depending on obstacles.
  - Verify that the student says "obstacle cleared" if there is an obstacle or simulated obstacle.
  - Verify that the student says positive rate of climb before retracting the flaps.

Soft-Field Approach and Landing

- Student does, student says:
  - Verify correct procedure.
  - Verify minimal braking.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 130.
- Assign homework:
  - Plan a cross country (preferably the three-leg cross-country that will be performed solo in the next stage). For example: KMLB-KSUA-KVRB-KMLB or KMLB-KOBE-KSUA-KMLB.
  - Review any deficiencies from this lesson.

Completion Standards

This lesson is complete when the student is able to, before the flight:

1. Have a completed navigation log prepared, with no errors.
2. Have the correct climb, cruise and descent performance calculations prepared.
3. Obtain an official weather briefing and explain how the existing weather, NOTAMs and conditions will affect the flight.

This lesson is complete when the student, during the flight, is able to (with little instructor guidance):

4. Intercept the planned course by using an on-course checkpoint.
5. Have predetermined ETAs accurate to ±6min.
6. Calculate ground speed ± 10kts based on actual time between checkpoints.
7. Maintains altitude ±200 ft from desired altitude and ±15° from desired heading.
8. Use radio navigation, pilotage and dead reckoning procedures.

This lesson is complete when the student, during the flight, is able to (without instructor guidance):

9. Conduct all normal, abnormal and emergency checklist procedures.